

Eight steps to shared Internet

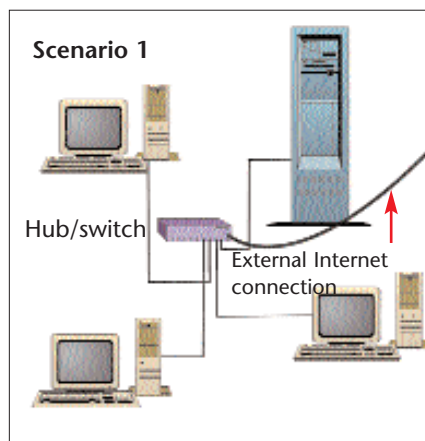
We used SyGate to set up an Internet server in eight easy steps

STEP 1 While installing SyGate, you need to remember that there's installation for both the server side, and for the client side. After installing the client and the server, the rest of the installation is a breeze. In the last step, you can manually configure the software.

STEP 2 The settings here vary according to need. We considered a couple of scenarios.

Scenario 1: You have a broadband connection wired directly to a hub, and at the same time, the server machine—henceforth, the machine that would act as the Internet server—is also connected to the same switch. This is called Single NIC mode.

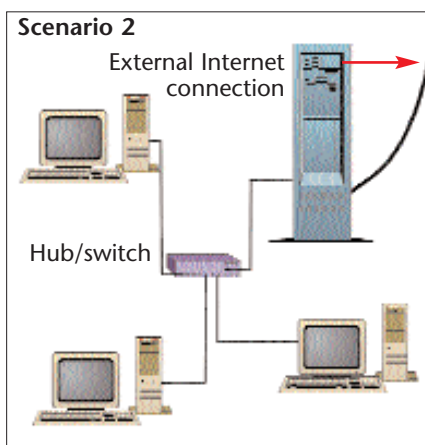
To establish Internet sharing, enable the settings for 'Use Single NIC mode'. As soon as you do this, you get a warning message. Click OK.



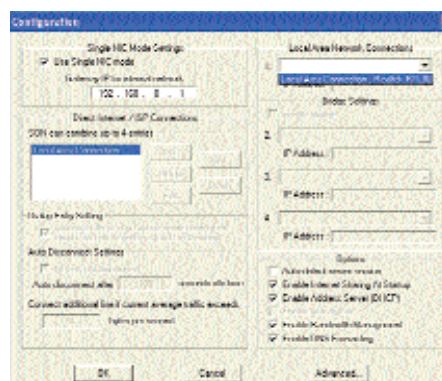
Scenario 2: If you have two separate connections, one each for connecting to the external network and the internal network, then keep the 'Use Single NIC mode' option unticked. SyGate will show the appropriate connections in 'Direct Internet/ISP Connections', as well as '1.' in 'Local Area Network Connection'.

The 'Direct Internet/ISP Connections' would have a 'Local Area Connection' if you have more than one LAN card or dial-up connection. To share a dial-up connection, you would first have to set up one in Windows, and then add it in SyGate.

STEP 3 Select the LAN card that connects your PC to the LAN in '1.' within 'Local Area Network Connection'.



STEP 4 Once Step 3 is complete, your configuration is done. But there are further settings to look at, for better optimisation of Internet bandwidth and system resources.



The configuration window of SyGate. The most common and the most important settings are specified here

If you have more than one connection, you need to specify the order in which they will be utilised, using the [Up] and [Down] keys. The ones appearing first will be used first, and so forth.

STEP 5 Dial-On-Demand and Auto Disconnect are two settings that will help you keep your dial-up costs to a minimum. If Dial-On-Demand is enabled, then each time a user wants to access the Internet, one of the connections will be activated depending upon precedence. Auto-disconnect will keep a tab on no-activity periods, and disconnect the call.

If you wish, you can configure the software to set up additional connections if bandwidth exceeds a certain threshold, by

enabling and specifying the threshold level for 'Connect additional line if...'

STEP 6 In the options section, 'Enable Internet Sharing At startup', 'Enable Activity Log', 'Enable Bandwidth Management' and 'Enable DNS forwarding' are recommended. Finally, if you have set up your network with a static IP for each PC, untick the 'Enable Address Server (DHCP)' option. A static IP means that no other machine in the world has the same IP as yours, which makes your machine uniquely addressable. If you've manually assigned IPs to each machine, you don't need a DHCP server.

STEP 7 You can modify some of the advanced settings for DHCP server or DNS forwarding by clicking on 'Advanced' in the configuration window.

If DHCP Server is enabled, you can specify the IP range that should be used while allocating IPs. This is particularly useful if some machines have static IPs and the others don't.

If DNS Forwarding is enabled, you need to specify one or more DNS server addresses for forwarding to work. Here, you can either use the DNS server your ISP has provided—use the ISP's helpline, or look up the documentation. Alternatively, you can use some of the most commonly used ones, such as those of VSNL—202.54.1.18 and 202.54.1.30. You need not specify any settings if you want to host a DNS server yourself.

STEP 8 The server-side setup, configuration and optimisation ends here. Now, you need to configure the clients to be able to use the server. On all machines that should have Internet connectivity, specify the gateway as the IP of the Internet server you just set up.

In Windows XP, go to *Start > Control Panel > Network Connections*. Here, right-click on the local area connection, and click Properties. In Properties, click 'Internet Protocol (TCP/IP)' from the list, and then again click Properties. Now specify the Default Gateway IP as that of the server IP.

Go to *Start > Settings > Control Panel > Network*. Here, click on 'TCP/IP', and click Properties. Go to the Gateway tab, and specify the server's IP.

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